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Advancing Diagnostic Equity through Incorporating Reflection on Implicit Bias into Clinical Reasoning Workshops

Lijia Xie, MD MS

Clinical Assistant Professor of Hospital Medicine, Stanford University

Mentor: Andre Kumar, MD, M.Ed

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Agenda

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Nothing to disclose

Overview of the Problem

Nearly 800,000 Americans experience permanent harm or death annually due to medical misdiagnosis (Newman-Toker *et al* 2024)

12 million Americans are misdiagnosed annually in clinics (Singh *et al* 2013)

For example, Black patients are up to 4 times more likely to experience a missed diagnosis for myocardial infarction compared to white counterparts (Herasevich *et al* 2023)

A meta-analysis of the impact of physician implicit bias on clinical decision making showed favoring of white patients, regardless of specialty, but the evidence for impact on decision making was low in quality (Dehon *et al* 2017)

No standardized curriculum on implicit bias recognition and management training for medical/PA students or residents

Diagnostic Equity

“We recommend the following definition of *diagnostic inequity*: the presence of preventable unwarranted variations in diagnostic processes among population groups that are socially, economically, demographically, or geographically disadvantaged.”

- Giardina *et al*, 2023

Women and people of color bear the brunt of medical misdiagnosis

Women and people of color are more likely than White men to experience a misdiagnosis, which can lead to deadly outcomes.



The Largest Health Disparity We Don't Talk About

Americans with serious mental illnesses die 15 to 30 years earlier than those without.

Why 'Unwell Women' Have Gone Misdiagnosed for Centuries

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A drawing of a hysterical woman, from a book by the French physician Paul Regnard,

Why Heart Disease in Women Is So Often Missed or Dismissed

New research shows that women may not realize their symptoms point to heart trouble, and that medical providers aren't picking up on it either.

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Research Aims

Aim 1a: To create a Diagnostic Equity (DE) curriculum that increases the amount of time learners spend discussing and reflecting on implicit bias during the clinical reasoning process, compared to learners receiving a standard clinical reasoning curriculum.

Aim 2: To evaluate the curriculum using a mixed-methods strategy, including pre- and post-intervention surveys and audio-recording of all workshops and post-intervention interviews for analysis of discussion times and key themes.

Aim 3: To provide a rewarding and safe experience for learners in the DE curriculum.

Short term goal: To achieve a greater degree of reflectiveness about implicit bias during the clinical reasoning process for medical and PA students and Family Medicine residents after a brief intervention.

Long-term goal: institutional adoption as a core curricular component of the IM rotation for third-year PA and medical students at Stanford and other medical schools through dissemination of the Diagnostic Equity curriculum, facilitation guide, and toolkit.

Research Methods: Study Design

Randomized controlled trial enrolling 3rd year medical and PA students at Stanford SOM and FM interns in small groups of 3-7 learners attending two 2-hour workshops

Control arm (n=15 students + 8 residents)

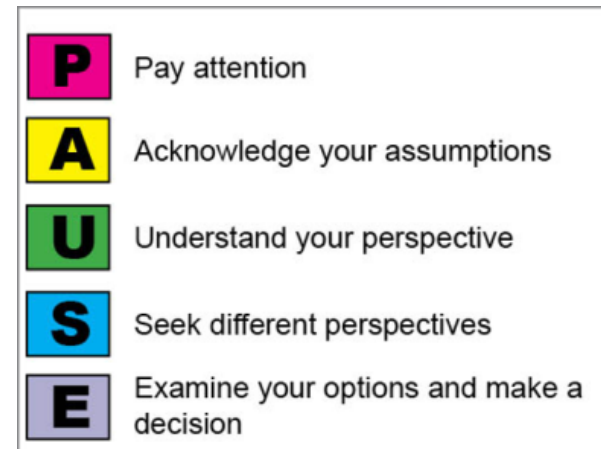
2 standard clinical reasoning workshops without implicit bias lecture and guided reflection

Case 1: young woman with chronic abdominal pain due to Wilson's disease

Case 2: incarcerated patient with bilateral lower extremity weakness due to COVID-induced transverse myelitis

Diagnostic Equity arm (n=15 students + 7 residents)

Diagnostic Equity didactic and structured reflection using PAUSE model, followed by 2 identical workshops



Curriculum & Worksheets

STEPS OF CLINICAL REASONING

1. DEFINE PROBLEM
2. USE A SCHEMA
3. MATCH CASE TO ILLNESS SCRIPT
4. COMPARE/CONTRAST DIAGNOSES
5. MAKE A COMMITMENT
6. COGNITIVE AUTOPSY

PROBLEM REPRESENTATION

A middle-aged man with HTN, HLD, previous smoker/OD, and recent COVID infection, presenting with acute onset of ascending bilateral numbness and leg pain, found to have BLE motor and sensory deficits with a sensory level and UMN signs with urinary retention.

IMPLICIT BIAS RECOGNITION AND MANAGEMENT

Pay attention: What is the situation and your reaction (physically, emotionally)?

Acknowledge your assumptions: What are your assumptions?

Understand your perspective: What is your perspective based on (past experiences, history)?

Seek other perspectives: What might be other perspectives?

Examine your options: What are you choosing to do?

How might implicit bias impact the care of this patient?

CAUSES OF CIRRHOSIS/PORTAL HTN

- Viral hepatitis
- Toxins (EtOH, herbal supplements, meds)
- NASH
- Autoimmune
- Genetic (hemochromatosis, Wilson's disease, alpha-1 antitrypsin deficiency)
- Primary biliary cirrhosis, primary sclerosis cholangitis
- Right-sided heart failure
- Infections (syphilis, brucellosis, echinococcus)

The Gendered Pain Gap: Examples

- National Hospital Ambulatory Medical Care Survey 2014 to 2018: 4152 young female patients (18–55 years) who presented to EDs with chest pain had 29% longer wait time compared with males, were less likely to have an EKG, be admitted to the hospital, or receive meds for acute coronary syndrome. Women of colour waited even longer and were less likely to be prescribed antiplatelet agents, narcotic analgesics, or benzodiazepines.
- UK Women's Health Survey of >110,000 women in 2021, 50% of respondents felt that their pain was disregarded or overlooked. "This dismissal was frequently accompanied with being informed that certain symptoms, particularly those related to menstrual health, were to be expected and accepted as a natural aspect of being a woman and thus did not deserve medical attention."
- Outcomes? Delays in diagnosis, poorly controlled pain, lower quality of life

Banco *et al.* "Sex and Race Differences in the Evaluation and Treatment of Young Adults Presenting to the Emergency Department With Chest Pain." *JAMA*, May 2022.

PRACTICE PROBLEM REPRESENTATION

How would you describe this case in one or two sentences?

Research Methods: Study Design

Quantitative analysis

Audio-record all workshops for participants in both groups.

Transcribe recordings, coded using a structured schema to identify instances of implicit bias discussion and calculate total time spent reflecting on implicit bias (during the case discussion time, not including the lecture and reflection exercise)

Compare the implicit bias discussion times between groups using median and IQR with box and whisker plot, Mann Whitney U test

Qualitative analysis

Pre- and post-intervention surveys to all participants, including 5-point Likert items and open-ended questions assessing knowledge, attitudes, and confidence around clinical reasoning and implicit bias recognition & management skills, and perceptions of workshop experience.

Semi-structured interviews

Thematic analysis with grounded theory of workshop transcripts and interview data

Research Questions

- 1) Among medical and PA students and FM residents, what is the difference in time spent reflecting on implicit bias during the clinical reasoning process between the group that receives a Diagnostic Equity curriculum vs standard clinical reasoning cases?
- 2) Among medical and PA students and FM residents, what is the level of confidence towards implicit bias recognition in the clinical reasoning process between the group that receives a Diagnostic Equity curriculum vs standard clinical reasoning cases?

Themes and quotes from DE workshops

Participants in the DE group acknowledge the risks of implicit bias in affecting patient care decisions and the possibility of making assumptions about patients' backgrounds.

"I definitely, in terms of **recognizing my own biases**, could see how both of those things like safety and then honesty of the patient would be **things that I probably would make assumptions about.**"

"The case presents with a young adult, with a history of anxiety, taking some medication for that, so, like, **an implicit bias might just be this patient might be either fully or exaggerating their symptoms or presentation**, and that might likely somewhat lead someone to either dismiss or not consider the full severity of the patient's presentation."

"Since she is a young woman with anxiety, **it can be passed off as, oh, this is just related to her menstrual cycle, or it's normal**, or I know it says in her social history that she hasn't had any recent sexual activity, but maybe they're just like, oh, maybe it's just something related to that."

"She might get dismissed very, very easily."

"On the topic of implicit bias, something that stood out to me was that one line in there where it said, that the patient had allegedly been in an altercation, and may or may not be trying to avoid going back to prison. Seems to imply that his concerns are not being treated as seriously or taken as seriously as it could be, or should be. **There's some sort of a hint of doubt on that side of the prison staff, that might impact his care.**"

Themes and representative quotes from the DE group

Participants in the DE group discuss the importance of continuing the workup based on relying on the significance of abnormal results.

“My first reaction was that he clearly had an abnormal physical exam, so treat that as it is with those things just completely separate and clearly, something is going on. So **it's our responsibility to just treat those findings like you would anyone else, even if you had assumptions**, acknowledging our assumptions, like our own bias.”

Participants in the DE group describe how they try to compartmentalize implicit bias in the clinical reasoning process and trust the patient's history.

“I said to assume the patient's telling the truth, and that they actually have a health problem until proven otherwise. I think that's kind of our moral responsibility as healthcare providers . . . there was the whoever, the guards or the somebody, talking about the patient had altercation last week, and trying to avoid going back to prison . . . having the altercation is something you hold in your mind. But you don't attach that to why they're here. It's just something you acknowledge, but just kind of here in a window. But then don't make any assumptions about that. And then **just take what the patient's giving you, assume it's the truth.”**

Themes and representative quotes

Preemptive reflection

“Before even going in to see a patient, maybe trying to recognize the certain things that I could foresee potentially having an implicit bias towards, so that it's on the forefront of my mind, and I'm aware of it going into a conversation. . . It's tough, because it's an implicit bias. You don't think you're biased towards certain things, and yet they can still be present. So it's definitely difficult to, you know, really train yourself to be aware of those things. So I think **my process would mostly be like taking a moment just beforehand to call out the things that I'm aware could potentially be influencing me, so that by verbalizing them, or acknowledging them, I can be more aware.**”

Awareness of the need for self-recognition

“One of the most helpful things for me is knowing what common implicit biases are, so that I can be sure of what not to do. That's the best way for me to address them, because they're sort of like implicit for a reason. So they exist. It's hard for me, and trying to identify our own, but knowing what some of the common ones are, is also super helpful in order to pinpoint where pitfalls may be for our own biases, so I think, knowing those and being aware of them, I definitely find myself already thinking about some of the things and some of the implicit biases of, you know, maybe patients with dementia or women, or different groups.”

Both groups: benefit of step-wise approach

“I liked the process of really kind of going through the schema to make sure you're including all the tests that you want. . . it was just a good reminder for me to, like, slow down and and really carefully and thoughtfully think through all the possibilities, because eventually you'll be the one ordering all the tests. **And if we don't order the tests, we won't get the answer right?**”

“I appreciate cases that look like something and are actually something else, because it just goes to show, It's a reminder that in medicine **it's important to not have biases until you have concrete answers, because so often we tend to assume that every patient or condition is going to be like an illness script that we learned.**”

Next Steps & Deliverables

Complete coding for implicit bias themes for analyzing difference in discussion times between DE and CR groups

Share findings with Stanford SOM leaders to incorporate into standardized 3rd year clinical reasoning curriculum

Complete Diagnostic Equity toolkit, including finalized workshop materials (handouts and slides), facilitator guides, and case content, for sharing on MedEdPORTAL



Lijia Xie, MD MS

Clinical Assistant Professor of Hospital Medicine

lxie2@stanford.edu

Thank you!

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